

7th Grade Unit 2 Assessment Guide

General Information	By the end of Grade 7, students are expected to perform all four operations with rational numbers with procedural fluency. Unit 2 builds upon the support for students' number sense from unit 1 by having students use different forms of a rational number in the setting of probability. The benchmarks MA.6.NSO.1.1 (Extend previous understanding of numbers to define rational numbers. Plot, order and compare rational numbers.) and MA.7.NSO.1.2 (Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals and percentages to solve mathematical and real-world problems.) are very similar. In Grade 6, students were to use number sense to plot, order, and compare rational numbers. In Grade 7, students build off that number sense to write equivalent forms of rational numbers, including repeating decimals. The benchmark MA.7.NSO.1.2 is an underlying benchmark for several other benchmarks in the course. The lessons in this unit both directly and indirectly involve knowledge for MA.7.NSO.1.2. Teachers should consider utilizing equivalent forms of rational numbers throughout the course. The general nature of probability encourages students to relate decimals, fractions, and percentages, which is why you find this benchmark within this unit. Teachers are encouraged not to back away from using rational numbers which, when written as a decimal, have one or two repeating digits. Although these are found on the unit assessment, it is a new topic for students, so they should continue to experience these numbers throughout the course.
Calculator Information	The questions on this assessment are calculator neutral. It is suggested that the teacher does not allow a calculator so that students are showing their knowledge of rewriting equivalent forms of rational numbers without the reliance on a calculator. As students progress in their number sense and fluency throughout the course, students can use a calculator for support.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1a	Benchmark(s)	MA.7.DP.2.1
{Pitcher, Batter}		Recommended Scoring (3 Total Points)	No partial credit should be given.
Question	1b	Benchmark(s)	MA.7.DP.2.1
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17}		Recommended Scoring (3 Total Points)	No partial credit should be given.
Question	2a	Benchmark(s)	MA.7.DP.2.3
10%		Recommended Scoring (4 Total Points)	No partial credit should be given.
Question	2b	Benchmark(s)	MA.7.DP.2.3
70%		Recommended Scoring (4 Total Points)	No partial credit should be given.

Question	3a	Benchmark(s)	MA.7.NSO.1.2				
<table><tr><th>Decimal</th><th>Percent</th></tr><tr><td>0.$\overline{54}$</td><td>54.$\overline{54}\%$</td></tr></table>		Decimal	Percent	0. $\overline{54}$	54. $\overline{54}\%$	Recommended Scoring (8 Total Points)	Consider giving 4 points for each equivalent form. Consider partial credit for students who show their work but make a minor error.
Decimal	Percent						
0. $\overline{54}$	54. $\overline{54}\%$						
Question	3b	Benchmark(s)	MA.7.NSO.1.2				
<table><tr><th>Fraction</th><th>Percent</th></tr><tr><td>$\frac{85}{100}$ or equivalent</td><td>85%</td></tr></table>		Fraction	Percent	$\frac{85}{100}$ or equivalent	85%	Recommended Scoring (8 Total Points)	Consider giving 4 points for each equivalent form.
Fraction	Percent						
$\frac{85}{100}$ or equivalent	85%						
Question	3c	Benchmark(s)	MA.7.NSO.1.2				
<table><tr><th>Decimal</th><th>Percent</th></tr><tr><td>$\frac{72}{100}$ or equivalent</td><td>0.72</td></tr></table>		Decimal	Percent	$\frac{72}{100}$ or equivalent	0.72	Recommended Scoring (8 Total Points)	Consider giving 4 points for each equivalent form.
Decimal	Percent						
$\frac{72}{100}$ or equivalent	0.72						

Question		4			Benchmark(s)	MA.7.DP.2.4
	Simulations				Recommended Scoring (15 Total Points)	Consider giving students 3 points for each correct selection. Consider deducting 2 points for each incorrect selection.
	A deck of 16 cards with 16 different letters.	A fair spinner with 7 equal sections where each is a different color.	A 10-sided die with the numbers 1 through 10.			
	A baseball player with an on-base percentage of 70%.	☐	☐			
	A basketball player making $\frac{5}{7}$ free throws.	☐	☒			
	A soccer goalie who stops $\frac{1}{2}$ of shots on the goal.	☒	☐			
	A hockey player with a shooting percentage of 25%.	☒	☐			

Question	5	Benchmark(s)	MA.7.DP.2.4
To use the ○ 10-sided die ○ deck of 16 cards ○ fair spinner with 7 equal sections to simulate the ○ soccer ○ hockey ○ baseball ○ basketball context 5 ○ cards ○ numbers ○ equal sections should be assigned to ○ being on base. ○ making a free throw. ○ stopping a shot on goal. ○ shooting a goal in hockey. OR To use the ○ 10-sided die ○ deck of 16 cards ○ fair spinner with 7 equal sections to simulate the ○ soccer ○ hockey ○ baseball ○ basketball context 5 ○ cards ○ numbers ○ equal sections should be assigned to ○ being on base. ○ making a free throw. ○ stopping a shot on goal. ○ shooting a goal in hockey.		Recommended Scoring (3 Total Points)	No partial credit should be given. A student should have a correctly completed statement to receive all of the points.

Question	6a	Benchmark(s)	MA.7.DP.2.3
	$\frac{13}{21}$	Recommended Scoring (4 Total Points)	No partial credit should be given.
Question	6b	Benchmark(s)	MA.7.DP.2.3
	$\frac{18}{21}$ or $\frac{6}{7}$	Recommended Scoring (4 Total Points)	No partial credit should be given.
Question	6c	Benchmark(s)	MA.7.DP.2.3
	$\frac{2}{21}$	Recommended Scoring (4 Total Points)	No partial credit should be given.

Question	6d	Benchmark(s)	MA.7.DP.2.2
The event that is less likely to occur is The event that is most likely to occur is The probability of the less likely event is closest to	○ drawing a marble with the letter N from the jar. ○ drawing a marble with the letter R or K from the jar. ○ drawing a marble with the letter other than H from the jar. ○ drawing a marble with the letter N from the jar. ○ drawing a marble with the letter R or K from the jar. ○ drawing a marble with the letter other than H from the jar. ○ 0. ○ 1.	Recommended Scoring (11 Total Points)	Consider giving students 4 points each for the first two statements and 3 points for the last statement.
Question	7	Benchmark(s)	MA.7.DP.2.4
The number of spins in probability that is closer to the probability.	○ Aaliyah's ○ Xavier's simulation results in a(n) ○ experimental ○ theoretical ○ experimental ○ theoretical	Recommended Scoring (3 Total Points)	No partial credit should be given.

Question		8	Benchmark(s)	MA.7.DP.2.2; MA.7.NSO.1.2
Least to Greatest		$\frac{8}{11}$, 75%, 0.77, 9 out of 10	Recommended Scoring (3 Total Points)	No partial credit should be given.
Question		9a	Benchmark(s)	MA.7.NSO.1.2
Percentage	12%		Recommended Scoring (3 Total Points)	No partial credit should be given.
Question		9b	Benchmark(s)	MA.7.NSO.1.2
Decimal	$0.2\bar{6}$		Recommended Scoring (3 Total Points)	No partial credit should be given.
Question		9c	Benchmark(s)	MA.7.NSO.1.2
Fraction	$\frac{2}{9}$		Recommended Scoring (3 Total Points)	No partial credit should be given.

Question		9d	Benchmark(s)	MA.7.NSO.1.2
Fraction $\frac{43}{225}$			Recommended Scoring (3 Total Points)	No partial credit should be given.
Question		9e	Benchmark(s)	MA.7.NSO.1.2
Decimal 0.2			Recommended Scoring (3 Total Points)	No partial credit should be given.